

When does the default PING circuit start to sing?

exp083 · 17 August 2026 · Draft

Abstract

This experiment asks a deliberately basic question about the reusable default `snn.components.ping()` circuit: as homogeneous Poisson drive increases, does it pass through a reproducible gamma regime? One compiled graph is held fixed across every condition. Only the input rate changes. The result is therefore both a first scientific example for SNNLANG and a test of what its advertised defaults actually do—without tuning them until the plot becomes photogenic.

Methods

Fixed authored graph

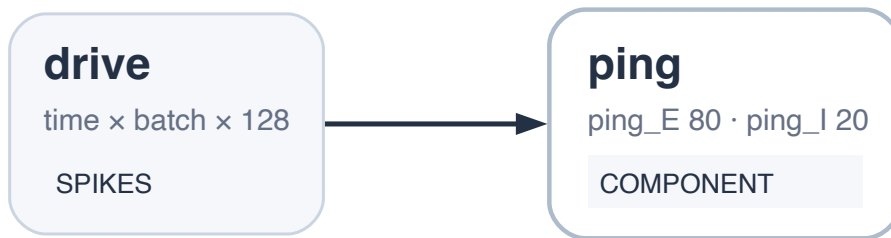


Figure 1: The single compiled graph reused at every drive condition.

A 128-channel typed spike input drives one default PING component containing 80 excitatory and 20 inhibitory cells. The timestep is 0.1 ms. Default neuron, synapse, delay, initializer, and constraint specifications are left untouched. The graph digest is `sha256:1173035a820aa4fb1c1770474cb465f78fc86d0f1523a52f17677d40c9ef6e3f`.

Registered drive sweep and measurements

Homogeneous Poisson drive spans 0 Hz, 25 Hz, 50 Hz, 75 Hz, 100 Hz, 125 Hz, 150 Hz, 200 Hz per input channel. Each condition contains 5 deterministic trials paired across rates, lasts 1000 ms, and discards the first 200 ms.

Gamma frequency uses the named `default-ping-gamma-v1` policy on the excitatory raster: a 30–80 Hz single-trial Welch spectrum, three-point sub-bin interpolation, rejection of band-edge maxima, and a peak-to-band-median power ratio greater than 3. Silence and failed quality gates remain unresolved rather than becoming 0 Hz. E–I lag is a separate descriptive cross-correlation measurement and does not decide whether gamma resolved.

Results

Exact responses at preselected drives

Fixed representative rates · trial seed 8300

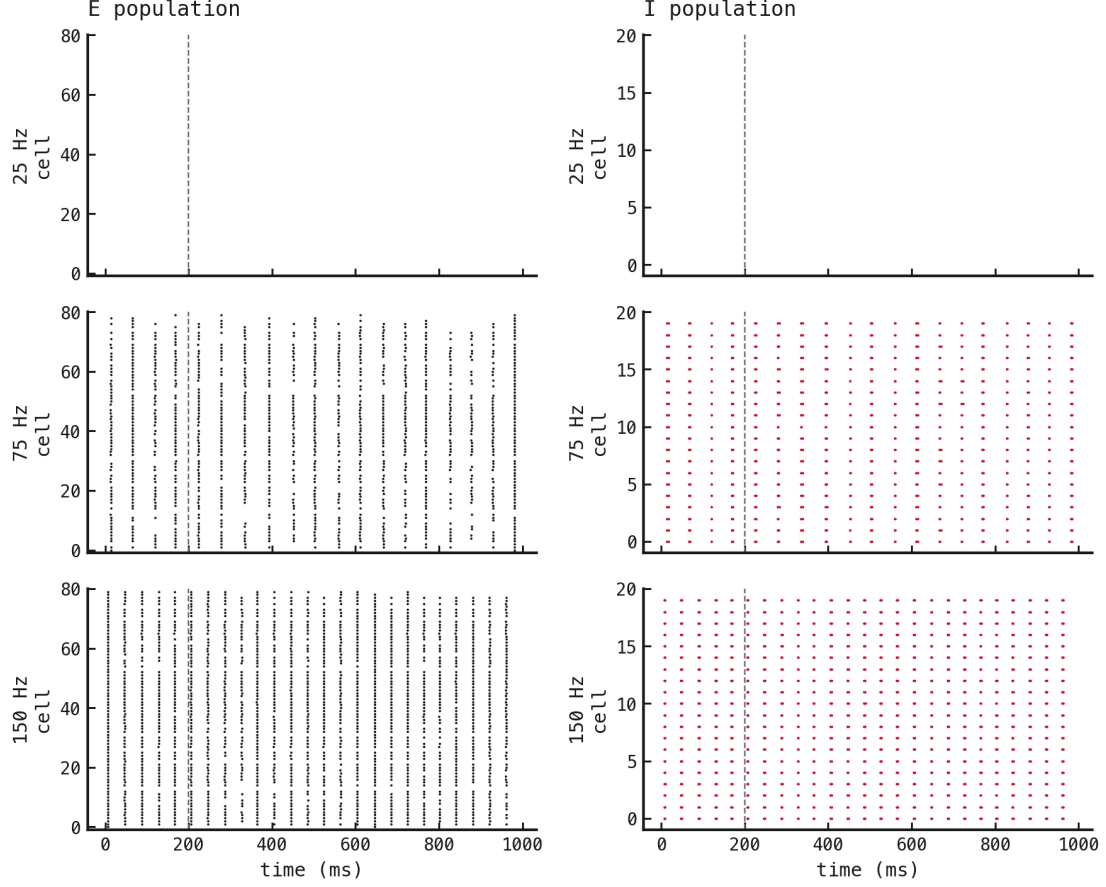


Figure 2: Exact E and I rasters at the three rates fixed before inspecting the primary sweep. The dashed line marks the end of burn-in.

Drive-response curve

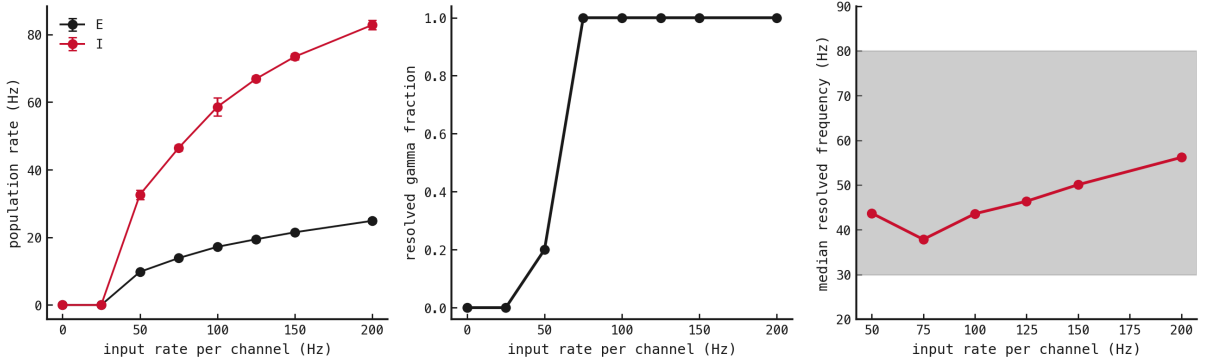


Figure 3: Population activation and the standardized gamma-frequency result across homogeneous drive. Error bars are standard deviations across paired trials.

The standardized estimator resolved at least one trial in 6 of 8 drive conditions. Resolution fraction, rather than the existence of a numerical spectral maximum, determines where the evidence supports rhythmic activity.

Spectral shape

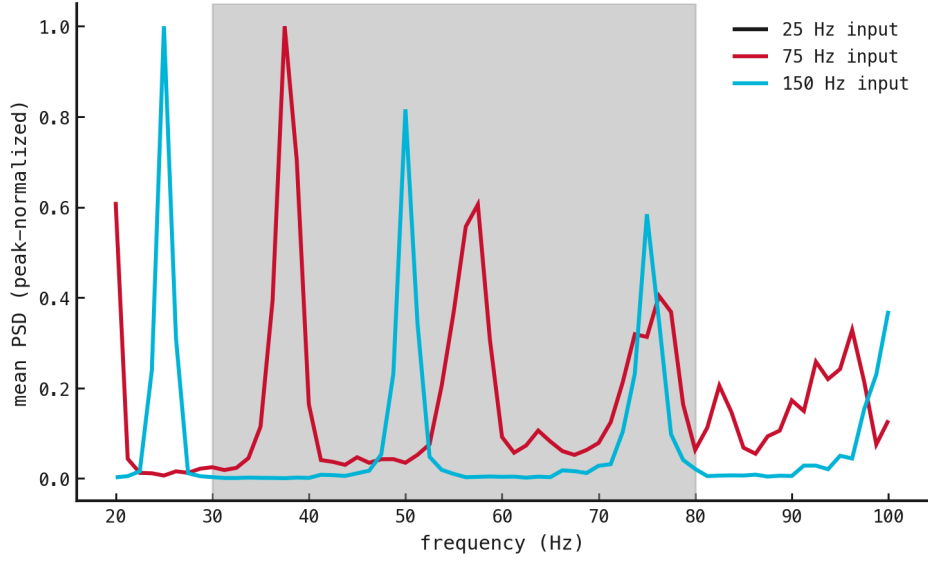


Figure 4: Mean E-population spectra at the preselected low, transitional, and high drives. Curves are peak-normalized for shape comparison; resolution uses unnormalized power.

Interpretation

The experiment is intentionally allowed to disappoint. A robust resolved interval supports the default component as a useful out-of-the-box PING regime. Activation without resolution says the defaults produce spikes but not convincing gamma under this protocol. Abrupt, seed-sensitive, or saturated responses instead identify where the reusable defaults need revision. All three outcomes are more useful than tuning in secret and publishing the prettiest survivor.